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Linear Programming Assignment: Optimal Number of Flights for Two Aircraft

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ABSTRACT An airline has two aircraft and wants to know how many flights each one should make. We model this as a linear programming problem and solve it with the graphical method and with the simplex method in MATLAB. The maximum benefit is 190 000 euro with 160 flights of A and 40 flights of B. The minimum fuel use is with 30 flights of each aircraft.

INDEX TERMS Graphical method, linear programming, MATLAB, simplex method.

I. INTRODUCTION

An airline has two aircraft, A and B, for one route. We want to know how many flights each aircraft should make to earn the most money. We write this as a linear programming (LP) problem and solve it.

II. PROBLEM FORMULATION

Let x_1 be the number of flights of aircraft A and x_2 the number of flights of aircraft B. The objective function is

$$\max Z = 1000\,x_1 + 750\,x_2. \tag{1}$$

The constraints are:

$$\begin{aligned} x_1 &\geq x_2 && \text{(C1)}\\ x_1 - x_2 &\leq 120 && \text{(C2)}\\ x_2 &\leq 180 && \text{(C3)}\\ x_1 + x_2 &\geq 60 && \text{(C4)}\\ x_1 + x_2 &\leq 200 && \text{(C5)}\\ x_1,\ x_2 &\geq 0 && \text{(nonnegativity)} \end{aligned}$$

For C1 we use $x_1 \geq x_2$ instead of $x_1 > x_2$, because LP does not work with strict inequalities.

III. PART 1A: GRAPHICAL SOLUTION

We draw every constraint as a line (Fig. 1). The green area is the feasible region. Then we compute Z in each extreme point (Table 1).

Z is highest in point C. So the best solution is $x_1 = 160$ flights for A and $x_2 = 40$ flights for B, which gives $Z = 190\,000$ €.

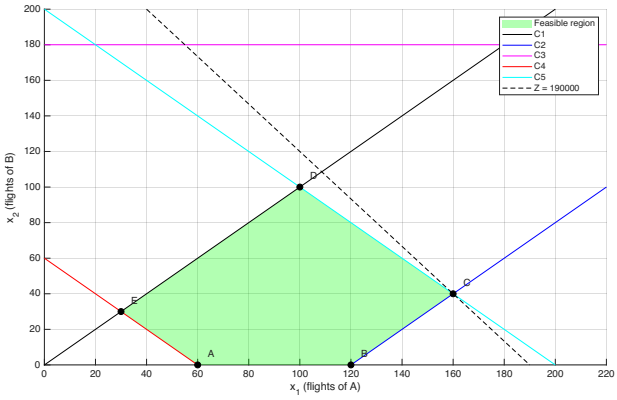


FIGURE 1. Graphical solution. The dashed line is the isoprofit line through the optimum.

TABLE 1. Extreme points and value of Z.

Point	Intersection	(x_1, x_2)	Z (€)
A	C4 and $x_2 = 0$	$(60, 0)$	60 000
B	C2 and $x_2 = 0$	$(120, 0)$	120 000
C	C2 and C5	$(160, 40)$	**190 000**
D	C1 and C5	$(100, 100)$	175 000
E	C1 and C4	$(30, 30)$	52 500

IV. PART 1B: REDUNDANT CONSTRAINTS

Yes. Constraint C3 ($x_2 \leq 180$) is a redundant constraint, because it does not change the feasible region. From C1 and C5 we get $2x_2 \leq x_1 + x_2 \leq 200$, so $x_2 \leq 100$ in the whole feasible region. In Fig. 1 the line of C3 lies above the green area. If we remove C3, the feasible region and the solution stay the same.

C1 and C4 are also not used at the optimum (point C is on C2 and C5). If we remove one or both of them, the feasible region gets bigger, but point C is still the best point. Only C2 and C5 cannot be removed, because they define point C.

V. PART 1C: SIMPLEX METHOD WITH MATLAB

MATLAB solves LP problems with `linprog`, in the form

$$\min_x f^T x \quad \text{subject to} \quad Ax \le b, \;\; x \ge lb.$$

Our problem is a maximization, so we minimize $-Z$:

$$\min \; -Z = -1000\,x_1 - 750\,x_2.$$

All constraints are written as $\le$. For C1 and C4 we multiply by -1:

$$A = \begin{bmatrix} -1 & 1 \\ 1 & -1 \\ 0 & 1 \\ -1 & -1 \\ 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 0 \\ 120 \\ 180 \\ -60 \\ 200 \end{bmatrix},$$

$$f = \begin{bmatrix} -1000 \\ -750 \end{bmatrix}, \quad lb = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

We use the simplex algorithm of `linprog`:

```
options = optimoptions('linprog', ...
          'Algorithm','dual-simplex');
[x,fval] = linprog(f,A,b,[],[],lb,[],options)
Z = -fval
```

MATLAB gives:

```
Optimal solution found.
x =
   160
    40
fval =
     -190000
Z =
      190000
```

So $x_1 = 160$, $x_2 = 40$ and $Z = -\text{fval} = 190\,000$ €. This is the same result as with the graphical method (point C).

VI. PART 2: MINIMUM FUEL CONSUMPTION

Let one flight of B use 1 unit of fuel. Then one flight of A uses 2 units. The constraints are the same as in Part 1, but now we minimize the total fuel:

$$\min F = 2\,x_1 + x_2.$$

Graphical method. The feasible region is the same as in Part 1 (Fig. 2). We compute F in each extreme point (Table 2).

F is lowest in point E, so $x_1 = 30$, $x_2 = 30$ and $F = 90$ units.

Simplex method with MATLAB. We use the same A, b and lb as in Part 1c, with $f = [2 \;\; 1]^T$. MATLAB gives:

```
Optimal solution found.
x =
    30
    30
fval =
    90
```

This is the same result as with the graphical method.

Note. In point E both aircraft make 30 flights, so A does not fly more than B. If A must make at least one flight more ($x_1 \ge x_2 + 1$), the best whole numbers are $x_1 = 31$ and $x_2 = 29$, with $F = 91$ units.

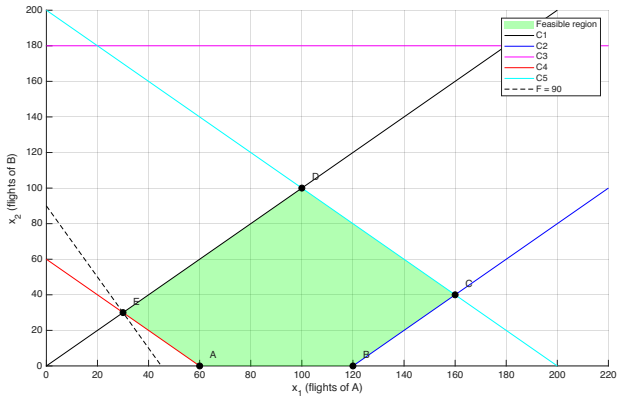


FIGURE 2. Graphical solution of Part 2. The dashed line is the isocost line through the optimum.

TABLE 2. Extreme points and value of F.

Point	(x_1, x_2)	F (units)
A	$(60, 0)$	120
B	$(120, 0)$	240
C	$(160, 40)$	360
D	$(100, 100)$	300
E	$(30, 30)$	**90**

VII. CONCLUSION

The maximum benefit is $Z = 190\,000$ € with $x_1 = 160$ and $x_2 = 40$. The graphical method and the simplex method in MATLAB give the same result. Constraint C3 is redundant. The minimum fuel use is $F = 90$ units with $x_1 = x_2 = 30$ (or $F = 91$ with $x_1 = 31$, $x_2 = 29$ if A must fly strictly more than B).